Numerical and simulation techniques in finance

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7.1 Variance reduction: general techniques

Antithetic variables

Let $X_1, X_2, \ldots, X_M$ be M identically distributed random variables (but not necessarily independent). If all pairs (X_i, X_j), $i \neq j$ have the same covariance $\mathbb{C}\mathrm{ov}(X_i, X_j)$, then

$$\mathbb{V}\mathrm{ar}\left[\frac{1}{M}\sum_{m=1}^{M} X_m\right] = \frac{1}{M}\mathbb{V}\mathrm{ar}\left[X\right] + \frac{M-1}{M}\mathbb{C}\mathrm{ov}(X_i, X_j)$$

The variance of the sample mean could be decreased if the random variables had negative covariance. If the distribution of X is symmetric then one can construct identically distributed simulations that are perfectly negatively correlated, known as **antithetic variables**.

Antithetic variables

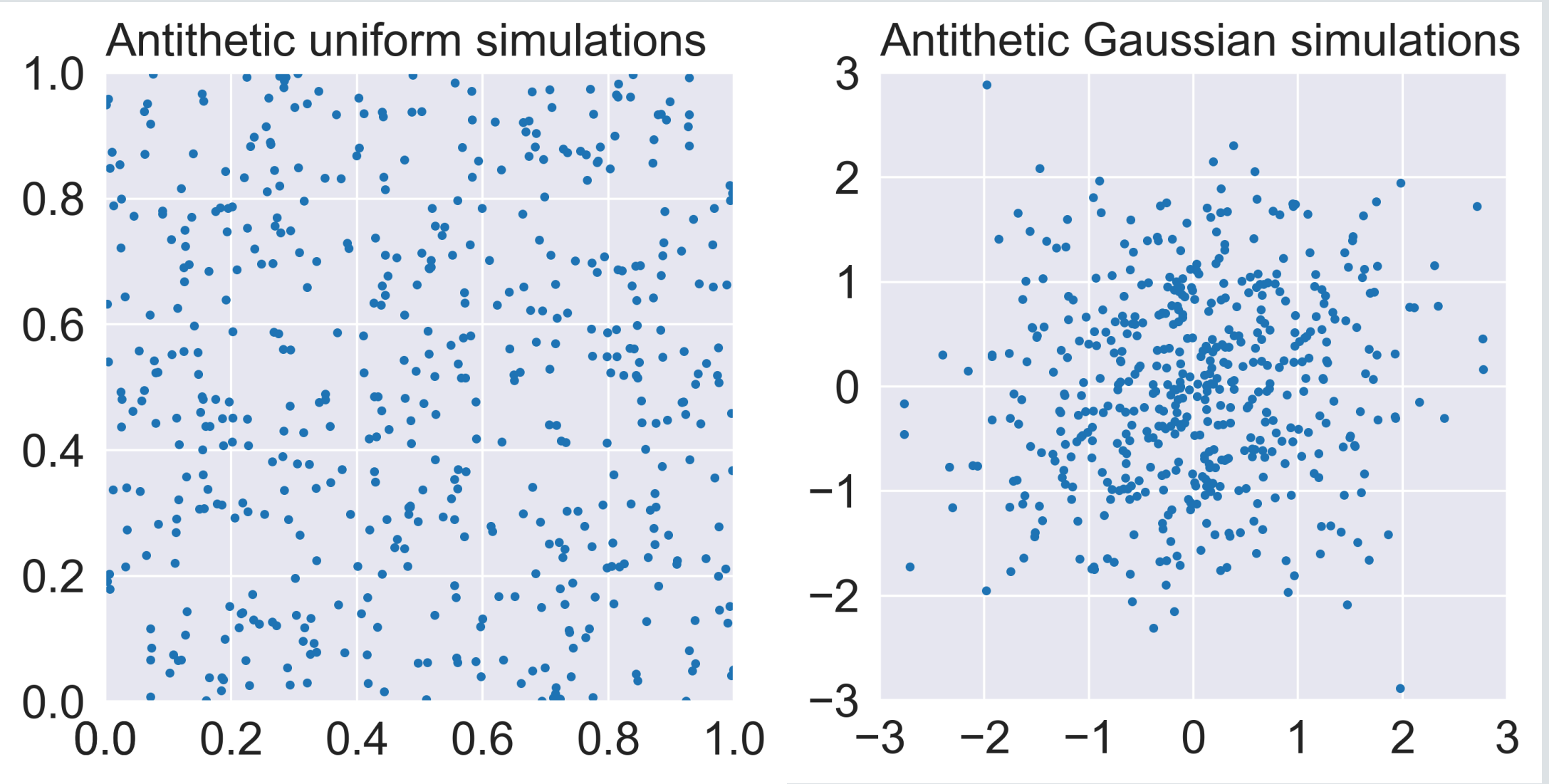
If $U \sim \mathcal{U}(0,1)$, then $1-U \sim \mathcal{U}(0,1)$ and $\mathbb{C}\mathrm{orr}(U, 1-U) = -1$

If $G \sim \mathcal{N}(\mu, \sigma^2)$, then $2\mu - G \sim \mathcal{N}(\mu, \sigma^2)$ and $\mathbb{C}\mathrm{orr}(G, 2\mu - G) = -1$

For each random variable X_m, let $\tilde{X}_m$ be its antithetic variable.
The **antithetic Monte Carlo estimator** of $\mathbb{E}[f(X)]$ for some function f is

$$\frac{1}{2}\left(\frac{1}{M}\sum_{m=1}^{M} f(X_m) + \frac{1}{M}\sum_{m=1}^{M} f(\tilde{X}_m)\right)$$

If f is monotonic, then the antithetic Monte Carlo estimator has lower variance than the classical Monte Carlo estimator.

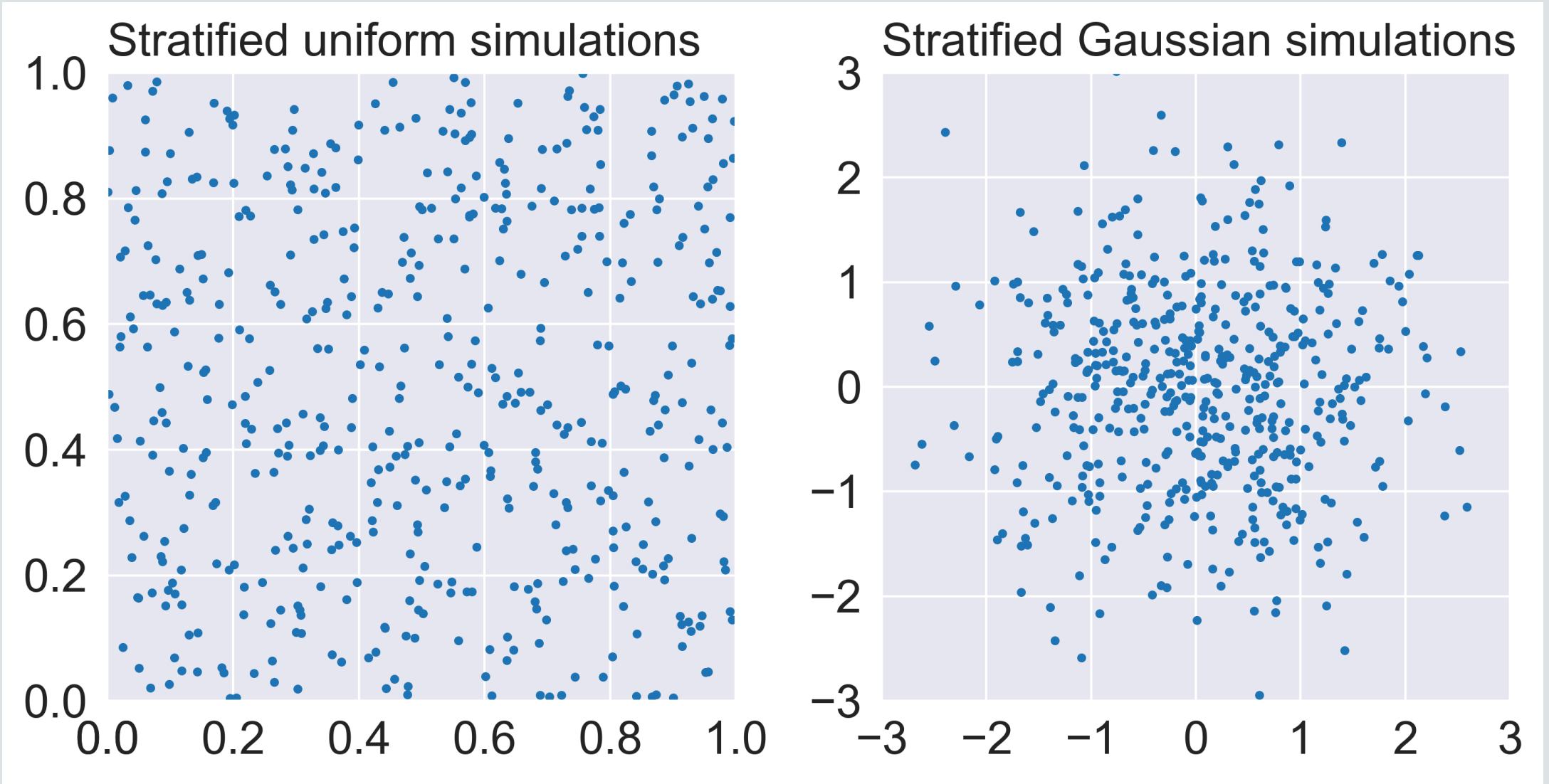

Antithetic uniform simulations
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Antithetic Gaussian simulations
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

Stratified sampling

Let $U \sim \mathcal{U}(0,1)$ be a standard uniform random variable. The idea of **stratified sampling** is to partition the $(0,1)$ interval into B segments of length $1/B$, and to draw M/B simulations in each segment.

For example, if $B = 5$, then the $(0,1)$ interval is split into five pieces $[0, 0.2)$, $[0.2, 0.4)$, $[0.4, 0.6)$, $[0.6, 0.8)$ and $[0.8, 1)$.
Then, in each interval $[u_b, u_{b+1})$, $b = 0, \ldots, 4$, we generate M/B simulations from a $\mathcal{U}(u_b, u_{b+1})$ uniform distribution.

The next slide shows $M = 500$ simulation of a bivariate standard uniform random variable stratified into $B = 25$ squares with equal area. Each small square contains exactly $M/B = 20$ simulations.


Stratified uniform simulations
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Stratified Gaussian simulations
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

Stratified sampling

More generally, let X be a random variable, and let Y be a discrete random variable with distribution $p_b = \mathbb{P}(Y = y_b)$, $b = 1, \ldots, B$. We know from the law of total expectation that

$$\mathbb{E}[f(X)] = \sum_{b=1}^{B} \mathbb{E}[f(X)|Y = y_b] \, \mathbb{P}(Y = y_b)$$

Stratified sampling can be applied by generating M_b simulations $X_{m,b}$, $m = 1, \ldots, M_b$ of the conditional random variable $X|\{Y = y_b\}$ for each $b = 1, \ldots, B$, compute the sample average $\overline{f}_b = \frac{1}{M_b} \sum_{m=1}^{M_b} f(X_{m,b})$, and finally compute the average $\sum_{b=1}^{B} p_b \overline{f}_b$.

Stratified sampling

A natural choice for M_b is $M_b := p_b M$. This is known as *proportional sampling*. If the conditional variances are known

$$\sigma_b^2 := \mathbb{V}\mathrm{ar}(f(X)|Y = y_b)$$

then the optimal variance reduction is obtained by defining M_b as

$$M_b := M \frac{p_b \sigma_b}{\sum_{j=1}^{B} p_j \sigma_j}$$

See section 3.3.3 of

[1] R. Korn, E. Korn, and G. Kroisandt. *Monte Carlo methods and models in finance and insurance*. CRC press, 2010.

Quasi Monte Carlo

If the simulations $U_1, U_2, \ldots, U_M$ are uniformly distributed in the hypercube $[0,1]^D$, then for any hypercube $H \subseteq [0,1]^D$, the proportion of points U_m inside H should be close to the volume of H:

$$\frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{U_m \in H\} \simeq \mathrm{Vol}(H)$$

The **discrepancy** of the sequence $\mathcal{U} := \{U_1, \ldots, U_M\}$ is defined as the worst absolute deviation between the two:

$$D_M(\mathcal{U}) := \sup_{H \subseteq [0,1]^D} \left| \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{U_m \in H\} - \mathrm{Vol}(H) \right|$$

Quasi Monte Carlo

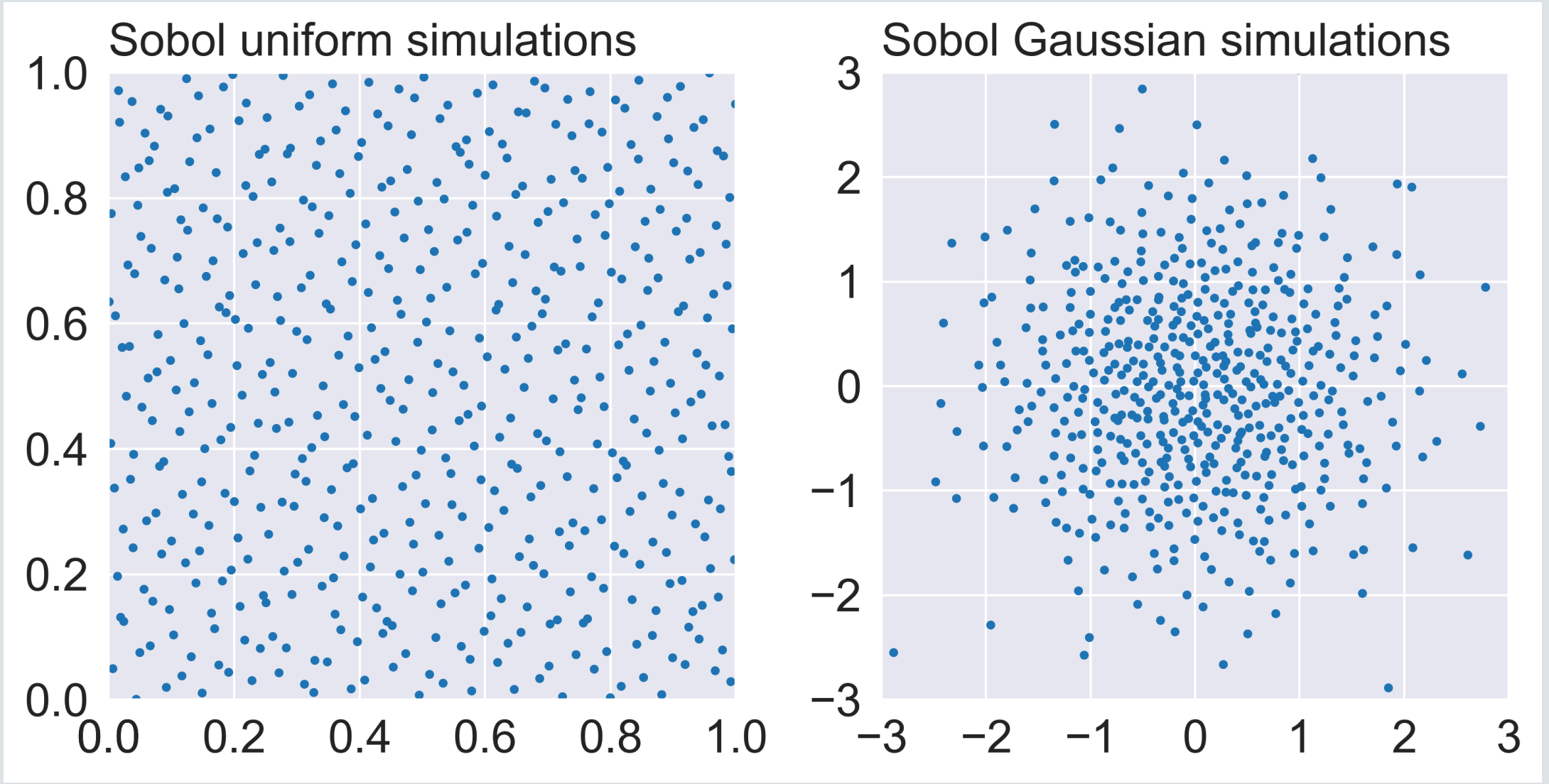
Ideally, the discrepancy should be as small as possible. A really random sequence has expected discrepancy

$$D_M(\mathcal{U}) = \mathcal{O}\left(\frac{\log(\log(M))}{\sqrt{M}}\right)$$

There exist deterministic **quasirandom** sequences, also known as **low discrepancy** sequences, with asymptotic discrepancy given by

$$D_M(\mathcal{U}) \underset{M\to\infty}{=} \mathcal{O}\left(\frac{\log(M)^D}{M}\right)$$

which is better than pseudorandom sequences for M large enough and/or D small enough. An example is given by **Sobol sequences**.


Sobol uniform simulations
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Sobol Gaussian simulations
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

Sobol sequences in Python

```
import matplotlib.pyplot as plt
from scipy.stats import qmc
sampler = qmc.Sobol(d=2, scramble=True)
sample = sampler.random_base2(m=9) # M=2**9=512 simulations
U1 = sample[:,0]
U2 = sample[:,1]

fig, ax = plt.subplots()
ax.scatter(U1,U2, s=30, marker=".", linewidth=0.35)
ax.set_xlim([0,1])
ax.set_ylim([0,1])
ax.set_xticks([0,0,.2,0.4,0.6,0.8,1])
ax.set_aspect('equal', 'box')
plt.title("Sobol uniform simulations",loc='left')
```

Optimal quantization

The idea of **optimal vector quantization** is to approximate a random variable $X \in \mathbb{R}^D$ by a discrete-valued random variable X^M taking M distinct values $x_1, x_2, \ldots, x_M$ with distribution $p_m = \mathbb{P}(X^M = x_m)$, $m = 1, \ldots, M$. The optimal values of x_m and p_m are obtained by minimizing some distance between X and X^M, for example

$$\min_{x_1, x_2, \ldots, x_M} \mathbb{E}\left[\min_{m=1,2,\ldots,M} L(X, x_m)\right]$$

where L is a loss function, for example $L(x, y) = \sum_{d=1}^{D} |x_d - y_d|^p$ for any two points $x \in \mathbb{R}^D$ and $y \in \mathbb{R}^D$.

Optimal quantization

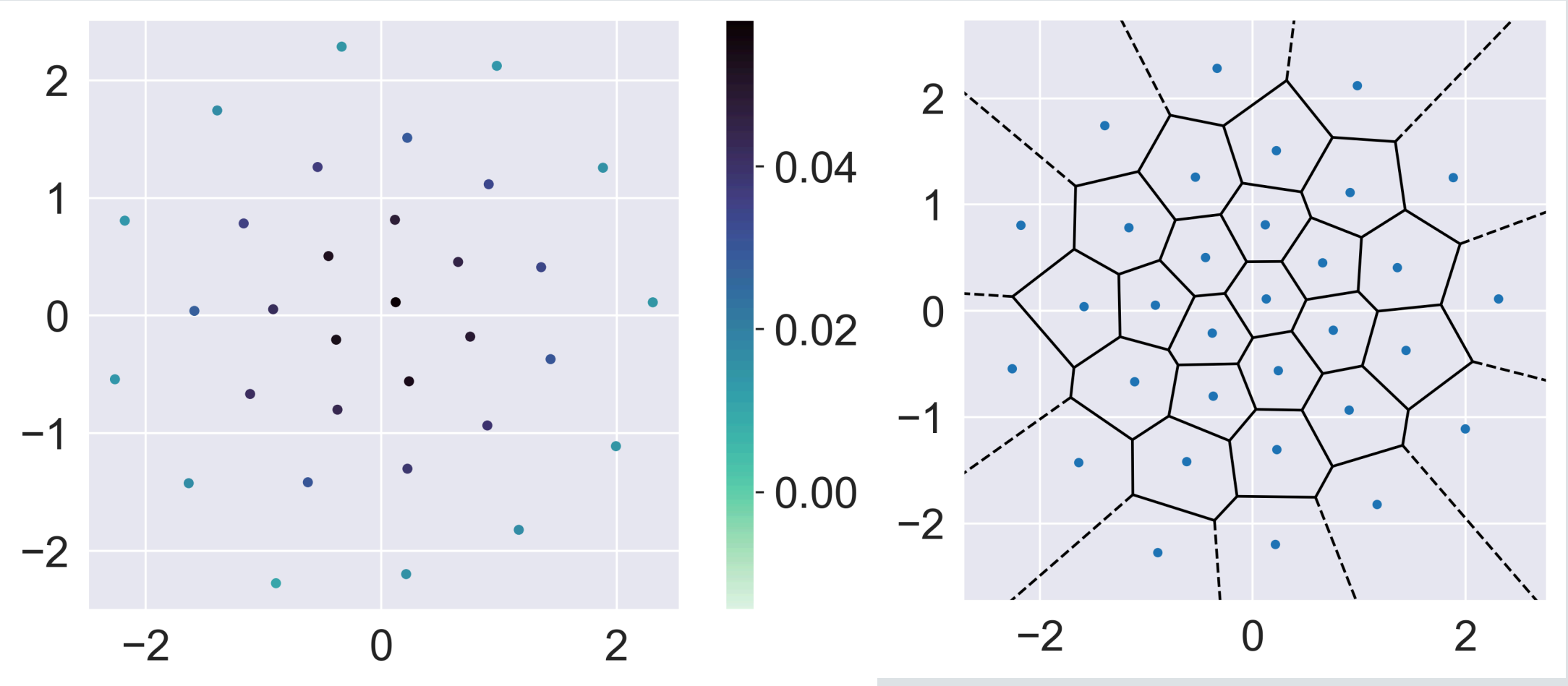
Using optimal quantization, the expectation $\mathbb{E}[f(X)]$ is approximated by the **weighted Monte Carlo estimator**

$$\mathbb{E}[f(X)] \simeq \sum_{m=1}^{M} p_m f(x_m)$$

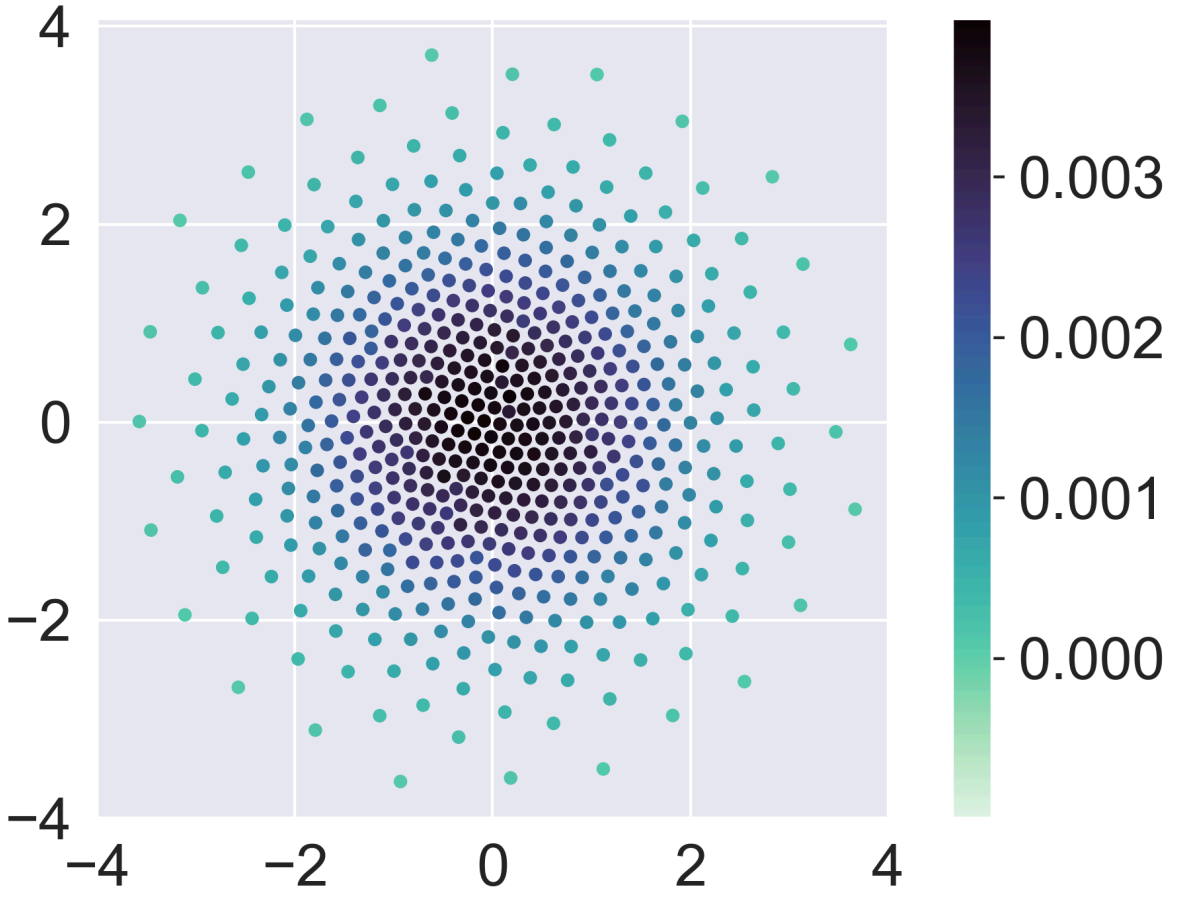
where the optimal quantization X^M of X has distribution $p_m = \mathbb{P}(X^M = x_m), m = 1, \ldots, M$.

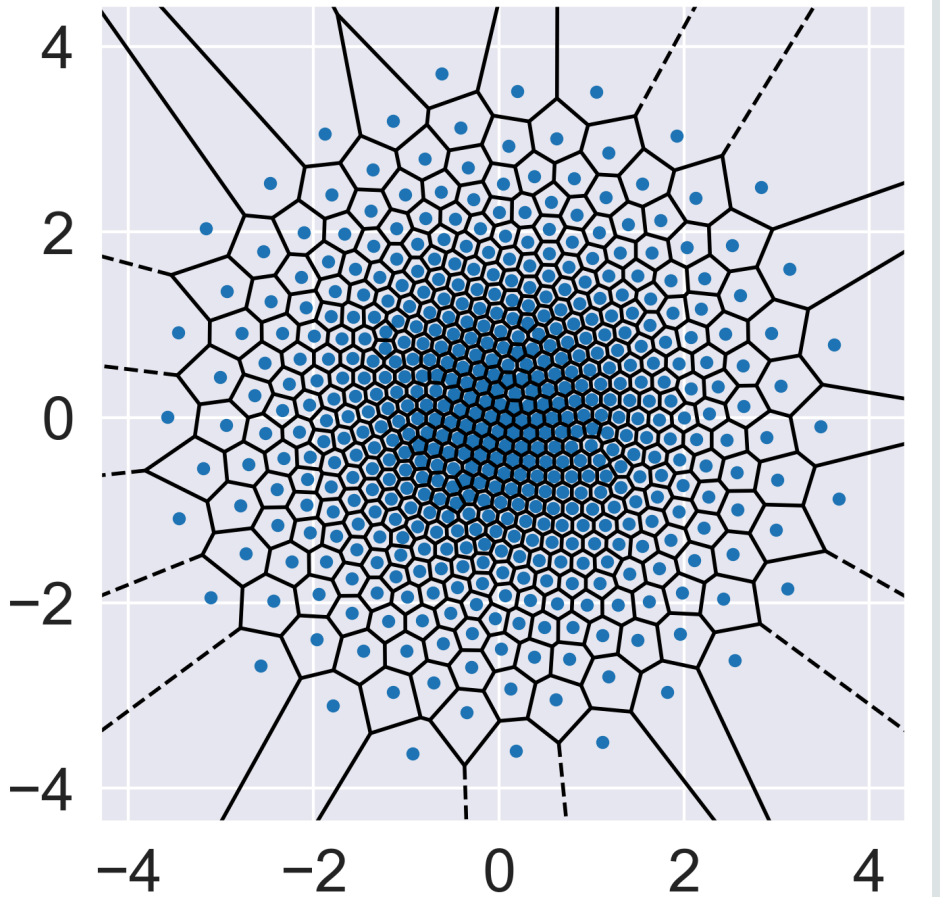
The optimal quantization grid is determined by the distribution of X, and needs to be precomputed. In the case of the Gaussian distribution, precomputed grids can be downloaded from http://quantize.maths-fi.com/ up to $M = 5999$ for $D = 1$, and up to $M = 1450$ for $2 \leq D \leq 10$.

M=32 Gaussian quantization points and their Voronoi diagram



M=512 Gaussian quantization points and their Voronoi diagram





⚠ Downside of non-independence

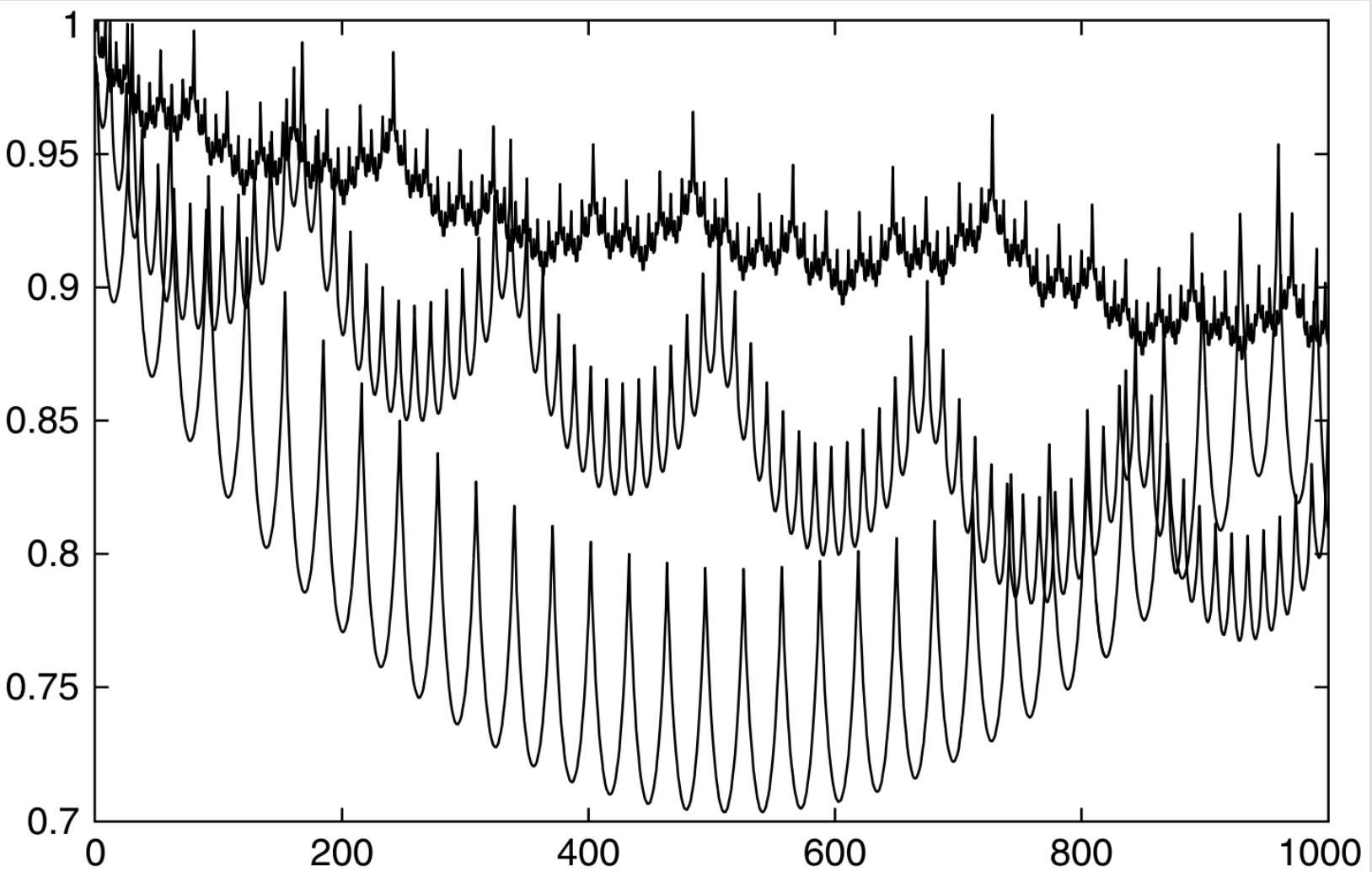
In all the methods discussed so far, the variance reduction is achieved by removing the *independence* between the simulations $X_1, \dots X_m$.

This means that such sequences cannot be used directly to simulate sample paths with independent increments (such as Brownian motions).

See section 9.3.4 of

[1] M. Gilli, D. Maringer, and E. Schumann. *Numerical methods and optimization in finance*. 2nd. Academic Press, 2019.

Downside of non-independence



Control variates

Let X be a random variable. We want to compute its expectation $\mathbb{E}[X]$. Suppose that we know another random variable Y which is "similar" to X, and for which we can compute $\mathbb{E}[Y]$ exactly. Then, Y can be used as a **control variate** in the Monte Carlo estimation of $\mathbb{E}[X] = \mathbb{E}[X - Y] + \mathbb{E}[Y]$

$$\hat{\mu}_{\mathrm{CV}}^{M} := \frac{1}{M} \sum_{m=1}^{M} (X_m - Y_m) + \mathbb{E}[Y]$$

where $(X_1, Y_1), \ldots (X_M, Y_M)$ are M i.i.d. simulations of (X, Y).
This control variate Monte Carlo estimator has a lower variance than the classical Monte Carlo estimator if $\mathbb{V}\mathrm{ar}(X - Y) < \mathbb{V}\mathrm{ar}(X)$.

Control variates

The control variate approach requires to find a suitable control variate Y on a case-by-case basis. The control variate must be such that

▷ $\mathbb{E}[Y]$ is explicit
▷ (X, Y) is easy to simulate
▷ $2\mathbb{C}\mathrm{ov}(X, Y) - \mathbb{V}\mathrm{ar}(Y)$ is large

See section 3.3.2 of

[1] R. Korn, E. Korn, and G. Kroisandt. *Monte Carlo methods and models in finance and insurance*. CRC press, 2010.

Multilevel Monte Carlo

Multilevel Monte Carlo is a generalization of the control variate idea. Let X_0, ..., X_{L-1} be a sequence of random variables which approximate X_L with increasing accuracy (but also increasing cost). We have the simple identity

$$\mathbb{E}[X_L] = \mathbb{E}[X_0] + \sum_{\ell=1}^{L} \mathbb{E}[X_\ell - X_{\ell-1}]$$

and the following unbiased estimator for $\mathbb{E}[X_L]$

$$\frac{1}{M_0} \sum_{m=1}^{M_0} X_{0,m} + \sum_{\ell=1}^{L} \left(\frac{1}{M_\ell} \sum_{m=1}^{M_\ell} (X_{\ell,m} - X_{\ell-1,m}) \right)$$

Multilevel Monte Carlo

$$\frac{1}{M_0}\sum_{m=1}^{M_0} X_{0,m} + \sum_{\ell=1}^{L}\left(\frac{1}{M_\ell}\sum_{m=1}^{M_\ell}(X_{\ell,m} - X_{\ell-1,m})\right)$$

where M_ℓ independent samples are used at each level ℓ of correction.
The optimal number of simulations M_ℓ can be obtained from the cost C_ℓ of computing one level ℓ correction sample $X_{\ell,m} - X_{\ell-1,m}$, and from the variance V_ℓ of the level ℓ correction $X_\ell - X_{\ell-1}$.

[1] M. Giles. "Multilevel Monte Carlo methods". In: *Acta Numerica* 24 (2015), pp. 259-328.

Importance sampling

Let X be a random variable, let X_1, X_2, ..., X_M be M i.i.d. copies of X, and let h be a real function. The classical Monte Carlo estimator of $\mu := \mathbb{E}[h(X)]$ is given by

$$\hat{\mu}^M := \frac{1}{M} \sum_{m=1}^{M} h(X_m)$$

Let f_X be the probability density function of X. We know that

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f_X(x) dx$$

Importance sampling

Let Y be another random variable with pdf $f_Y > 0$. Let Y_1, Y_2, ..., Y_M be M i.i.d. copies of Y, and let h be a real function. The expectation $\mu = \mathbb{E}[h(X)]$ can written as

$$\begin{aligned}
\mathbb{E}[h(X)] &= \int_{-\infty}^{\infty} h(x) f_X(x) dx \\
&= \int_{-\infty}^{\infty} h(x) \frac{f_X(x)}{f_Y(x)} f_Y(x) dx \\
&= \mathbb{E}\left[h(Y) \frac{f_X(Y)}{f_Y(Y)} \right]
\end{aligned}$$

Importance sampling

The two random variables $h(X)$ and $h(Y)f_X(Y)/f_Y(Y)$ have the same expectation, but don't necessarily have the same variance. With a clever choice of density f_Y, the random variable $h(Y)f_X(Y)/f_Y(Y)$ will have a lower variance than the original random variable $h(X)$. In such a case, the new Monte Carlo estimator

$$\hat{\mu}_{\mathrm{IS}}^{M} := \frac{1}{M}\sum_{m=1}^{M} h(Y_m)\frac{f_X(Y_m)}{f_Y(Y_m)}$$

will have a lower variance than the original Monte Carlo estimator $\hat{\mu}^M$. This is an example of **importance sampling** estimator. The ratio f_X/f_Y is called **likelihood ratio**, or **Radon-Nikodym derivative**.

Importance sampling example

Consider for example the Monte Carlo estimation of the expectation $\mathbb{E}[h(X)\mathbf{1}_{\{X \geq B\}}]$. The direct Monte Carlo estimator

$$\frac{1}{M}\sum_{m=1}^{M} h(X_m)\mathbf{1}_{\{X_m \geq B\}}$$

might generate many zero simulations due to the indicator function

$$\mathbf{1}_{\{X_m \geq B\}} = \begin{cases} 1 & \text{if } X_m \geq B \\ 0 & \text{if } X_m < B \end{cases}$$

Importance sampling example

Note that the expectation can be written as

$$\begin{aligned}\mathbb{E}[h(X)\mathbf{1}_{\{X\geq B\}}] &= \mathbb{E}[h(X)|X\geq B]\mathbb{P}(X\geq B)\\ &= \mathbb{E}[h(Y)](1-F_X(B))\end{aligned}$$

where Y is the conditional random variable $X|X\geq B$, and F_X is the CDF of X. We know that the random variable X can be simulated from the inversion method $X=F_X^{-1}(U)$ where U is a standard uniform random variable. The truncated distribution Y can be simulated by the truncated inversion method $Y=F_X^{-1}(U_B)$ where U_B is a uniform random variable on the interval $[F_X(B),1]$.

Importance sampling example

Then, the importance sampling estimator

$$\left(\frac{1}{M} \sum_{m=1}^{M} h(Y_m) \right) (1 - F_X(B))$$

will have a lower variance than the original Monte Carlo estimator, as it doesn't contain an indicator function anymore.

7.2 Put-Call parity

Variance reduction of Monte Carlo estimation of European Call price

Here is an example of variance reduction technique which is specific to option pricing, and more precisely to the pricing of Call options. The idea is based on:

▷ The Put-Call parity, which connects the price of a Call and a Put with the same strike to the price of a forward contract,
▷ The fact that the price of a forward contract is explicit,
▷ The fact that the variance of the Monte Carlo estimation of a Put price is lower than the one of a Call price.

Forward contract

A **forward contract** with delivery price K obligates its holder to buy one share of the stock S at expiration time T in exchange for payment K. At expiration, the value of the forward contract is $S_T - K$. By a **static hedging** argument, the value of the forward contract at time at time t is

$$F_t = S_t - e^{-r(T-t)}K$$

Generally, the delivery price K of the contract is chosen such that $F_0 = 0$, i.e. $K = S_0 e^{rT}$, called the *forward price*. See Chapter 4.5.6 in

[1] S. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.

Put-Call parity

The payoff of the forward contract is

$$F_T = S_T - K = (S_T - K)^+ - (K - S_T)^+$$

It is equal to a European Call option payoff minus a European Put option payoff. An arbitrage argument implies that at any time t, the value of the forward contract is equal to the price of a European Call option minus the price of a European Put option with strike price K

$$F_t = \mathrm{Call}_t(K, T) - \mathrm{Put}_t(K, T)$$

This formula is called the **Put-Call parity**

Put-Call parity

One can use the Put-Call parity to estimate the price of a Call option from the price of a Put option and a forward contract:

$$\mathrm{Call}_t(K,T) = F_t + \mathrm{Put}_t(K,T) = S_t - e^{-r(T-t)}K + \mathrm{Put}_t(K,T)$$

In particular, the Put-Call parity provides an easy estimation of the Call option price from the Monte Carlo estimation of a Put option price with the same strike price K.

In order to estimate the price of a Call option by Monte Carlo, it is actually better to first estimate the price of a Put option by Monte Carlo and then use the Put-Call parity. The reason for that is that the Monte Carlo estimator of a Put option **has lower variance** than the one of a Call option.

Bounded Put payoff

Indeed, the payoff $(K - S_T)^+ = \max(K - S_T, 0) \leq K$ of a Put option is bounded, which is not the case of the payoff of a Call option. This means that the price of a Put option is always bounded by K.

$$0 \leq \mathrm{Put}_t(K, T) = \mathbb{E}\big[e^{-r(T-t)}(K - S_T)^+\big] \leq K$$

which is also true of the Monte Carlo estimator of the Put option price

$$0 \leq \frac{1}{M} \sum_{m=1}^{M} e^{-r(T-t)}(K - S_T^m)^+ \leq K$$

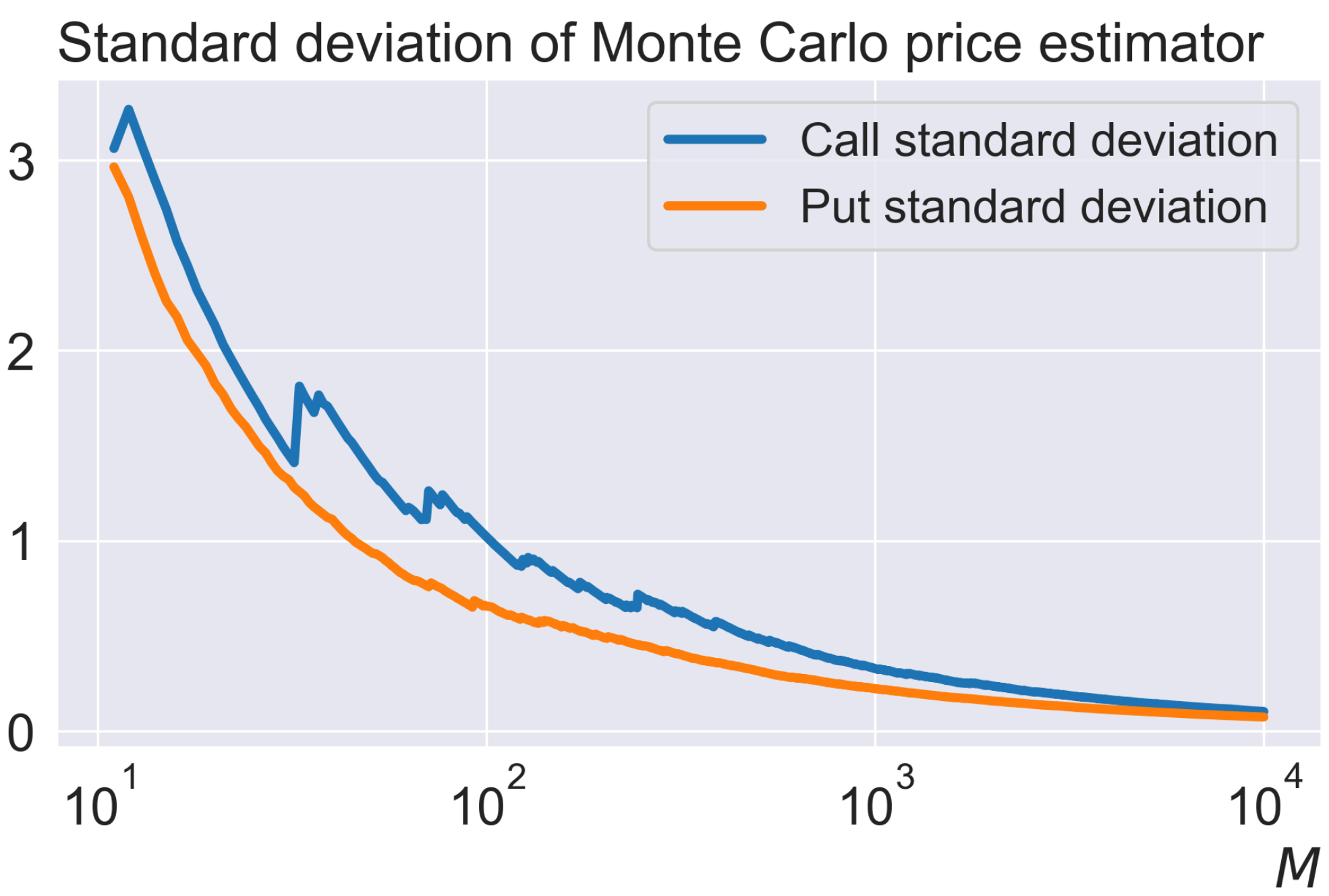
Bounded Put payoff

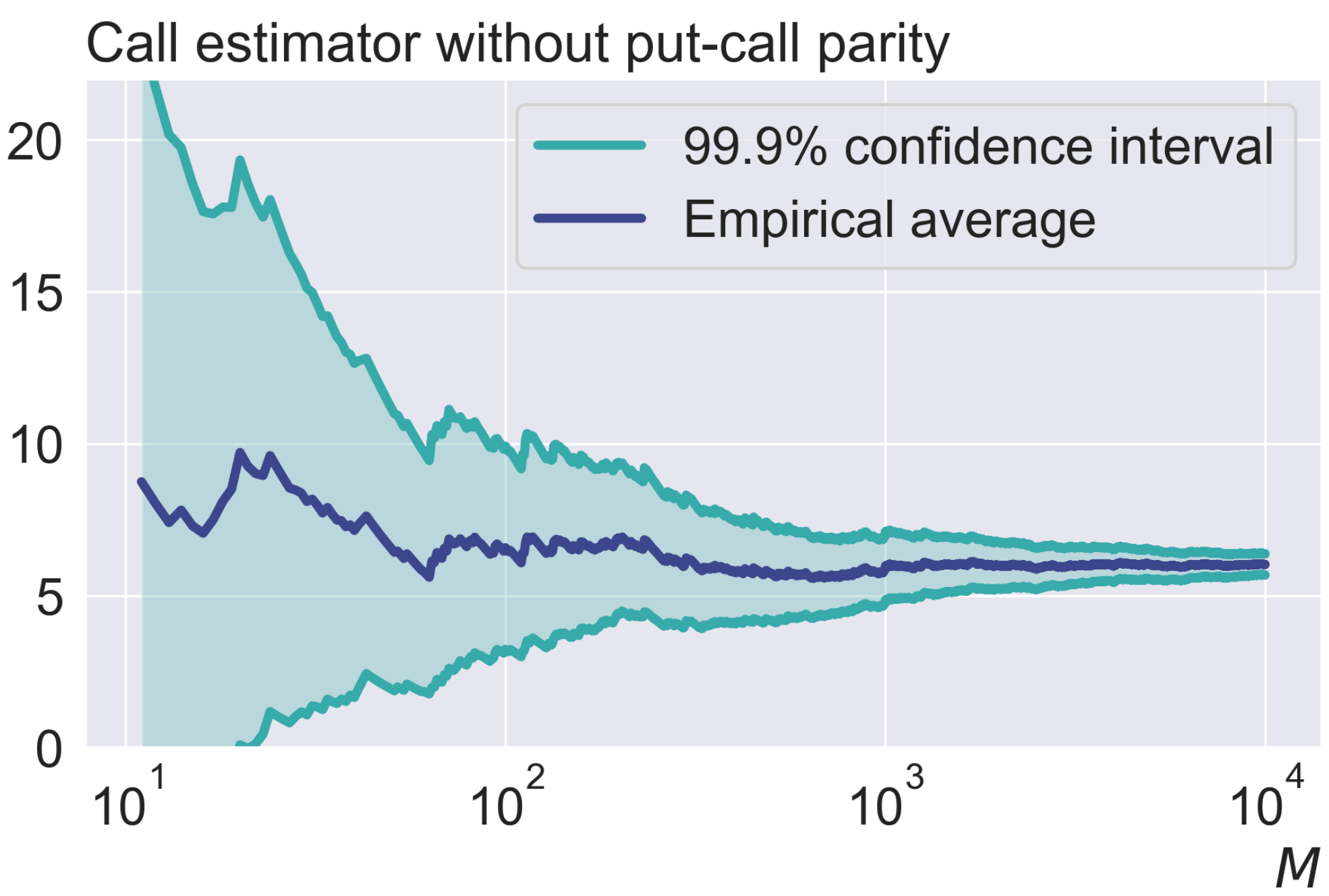
Since the Monte Carlo estimator of the Put option price is bounded, the same is true of the $\mathcal{L}_p$-norm of this estimator, for any $p > 0$:

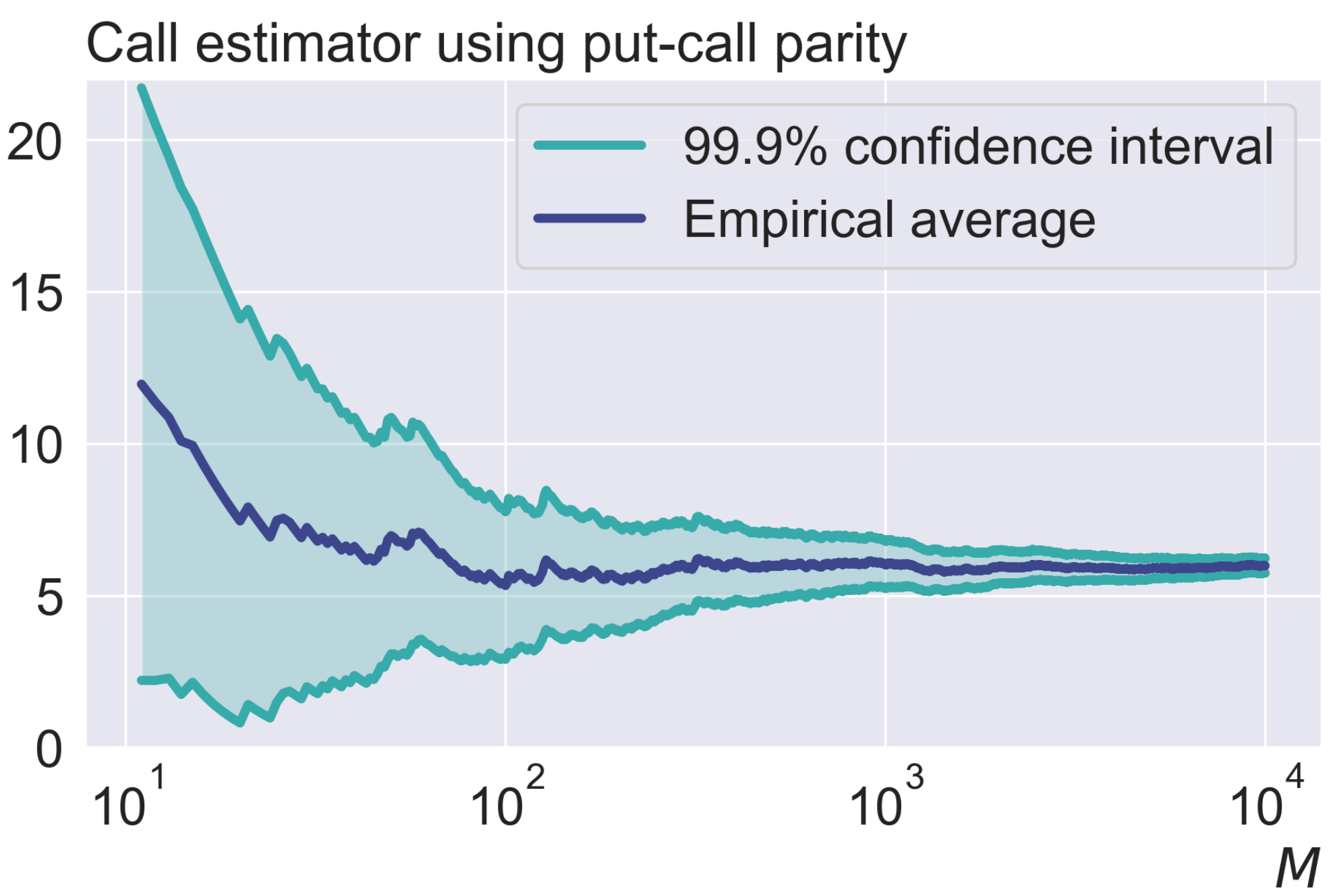
$$\left\| \frac{1}{M} \sum_{m=1}^{M} e^{-r(T-t)} (K - S_T^m)^+ \right\|_p = \mathbb{E}\left[\left(\frac{1}{M} \sum_{m=1}^{M} e^{-r(T-t)} (K - S_T^m)^+ \right)^p \right]^{1/p} \leq K$$

In particular all the moments of the Monte Carlo Put estimator are bounded. By contrast, the bounds on the moments of the Monte Carlo Call estimator are not that tight.

In the next slides, we consider the Monte Carlo estimation of a European Call option price in the Black-Scholes model, using the parameters $T = 1$, $N = 100$, $r = 0.05$, $\sigma = 0.3$, $S_0 = 50$, $K = S_0 e^{rT}$.


Standard deviation of Monte Carlo price estimator
Call standard deviation
Put standard deviation
3
2
1
0
10^1
10^2
10^3
10^4
M


Call estimator without put-call parity
99.9% confidence interval
Empirical average
20
15
10
5
0
10^1
10^2
10^3
10^4
M


Call estimator using put-call parity
99.9% confidence interval
Empirical average
20
15
10
5
0
10^1
10^2
10^3
10^4
M

7.3 Hedged Monte Carlo

Financial option pricing

The classical way to price options is to prescribe a **risk-neutral** stochastic model for the underlying asset price dynamics. This stochastic model under this **pricing measure** is (generally) different from the fitted stochastic model under the **historical measure** (based on the actual historical data).

For example, in the Black-Scholes model, the asset price S_t is assumed to follow a Geometric Brownian motion dynamics $dS_t = \mu S_t dt + \sigma S_t dW_t$ where μ is the drift parameter and σ is the volatility parameter. These are the actual parameters to fit on the historical price data. Let $\mathbb{P}$ denote this historical probability measure

Financial option pricing

Assuming a constant risk-free rate r, the premium of a European option with payoff f at time T is (generally) NOT equal to $e^{-rT}\,\mathbb{E}_{\mathbb{P}}[f(S_T)]$. Such a premium would make the average profit & loss $e^{-rT} f(S_T) - e^{-rT}\,\mathbb{E}_{\mathbb{P}}[f(S_T)]$ of a passive option seller, equal to zero on average, with possibly huge residual risk.

Instead, option sellers actively try to reduce their residual risk by buying and selling a carefully chosen amount of the underlying asset S over time. In the Black-Scholes model, the optimal dynamic hedging strategy exists and requires the premium of the option to be equal to $P_0 := e^{-rT}\,\mathbb{E}_{\mathbb{Q}}[f(S_T)]$, where the expectation is taken under the so-called *pricing measure* under which the asset price dynamics is given by $dS_t = rS_t dt + \sigma S_t dW_t$ (the drift μ is replaced by the risk-free rate r).

Financial option pricing

Moreover, the **optimal hedging strategy** is to maintain a portfolio containing Δ_t units of the underlying asset at time t, where $\Delta_t := e^{-r(T-t)} \frac{\partial \mathbb{E}_{\mathbb{Q}}[f(S_T)|S_t]}{\partial S_t}$.

In the Black-Scholes model, the terminal value (at time T) of a portfolio short of one option and long Δ_t unit of the underlying S with be exactly zero (i.e. the *residual risk* is zero). This is only true in the Black-Scholes model, and the risk-neutral pricing approach is only a mathematical trick to easily incorporate the hedging strategy in the Black-Scholes model. As soon as one of the assumptions of the Black-Scholes model is removed (for example if the portfolio is not rebalanced continuously, as in reality) then the residual risk is not zero anymore.

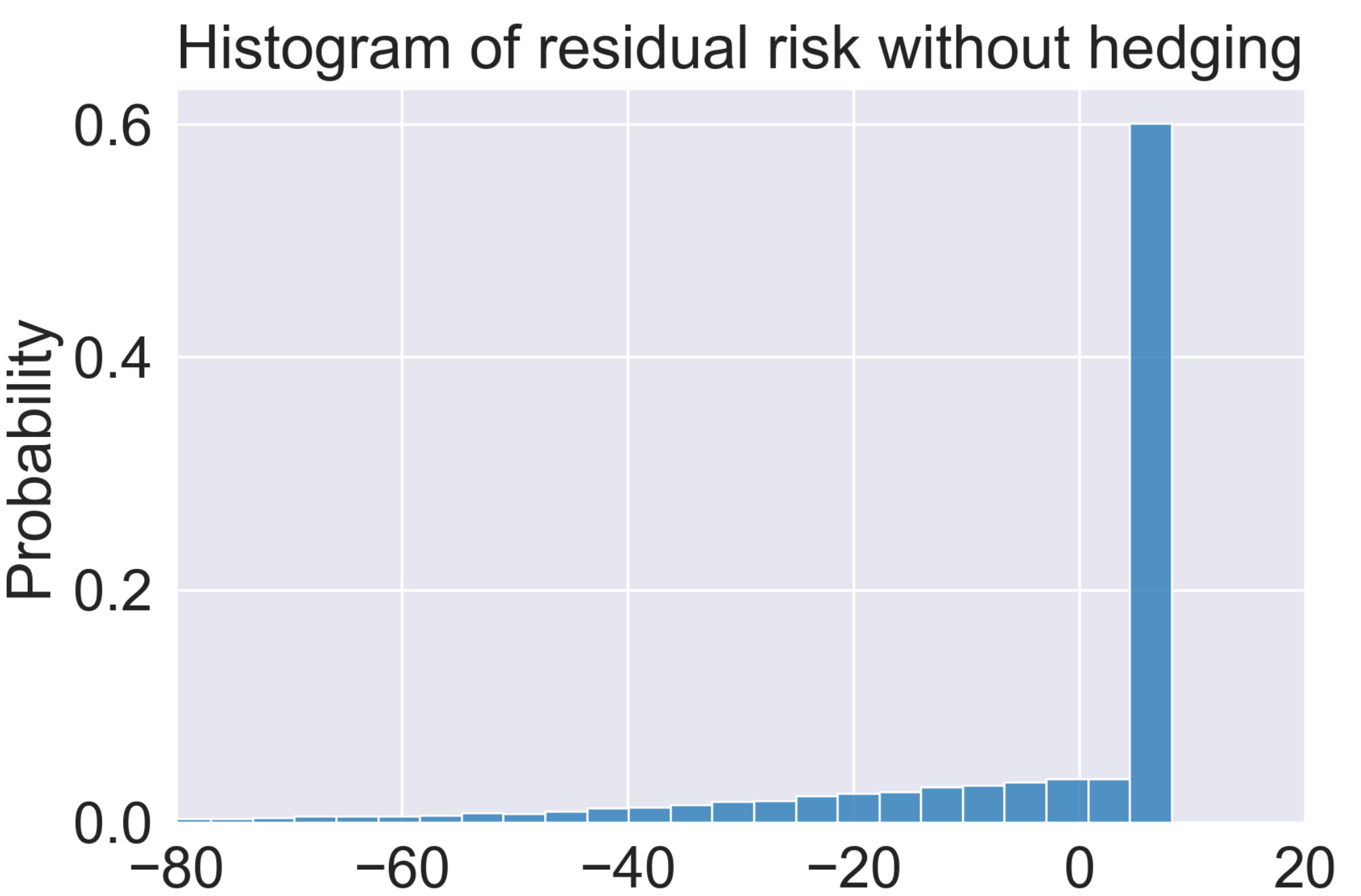
Hedged Monte Carlo

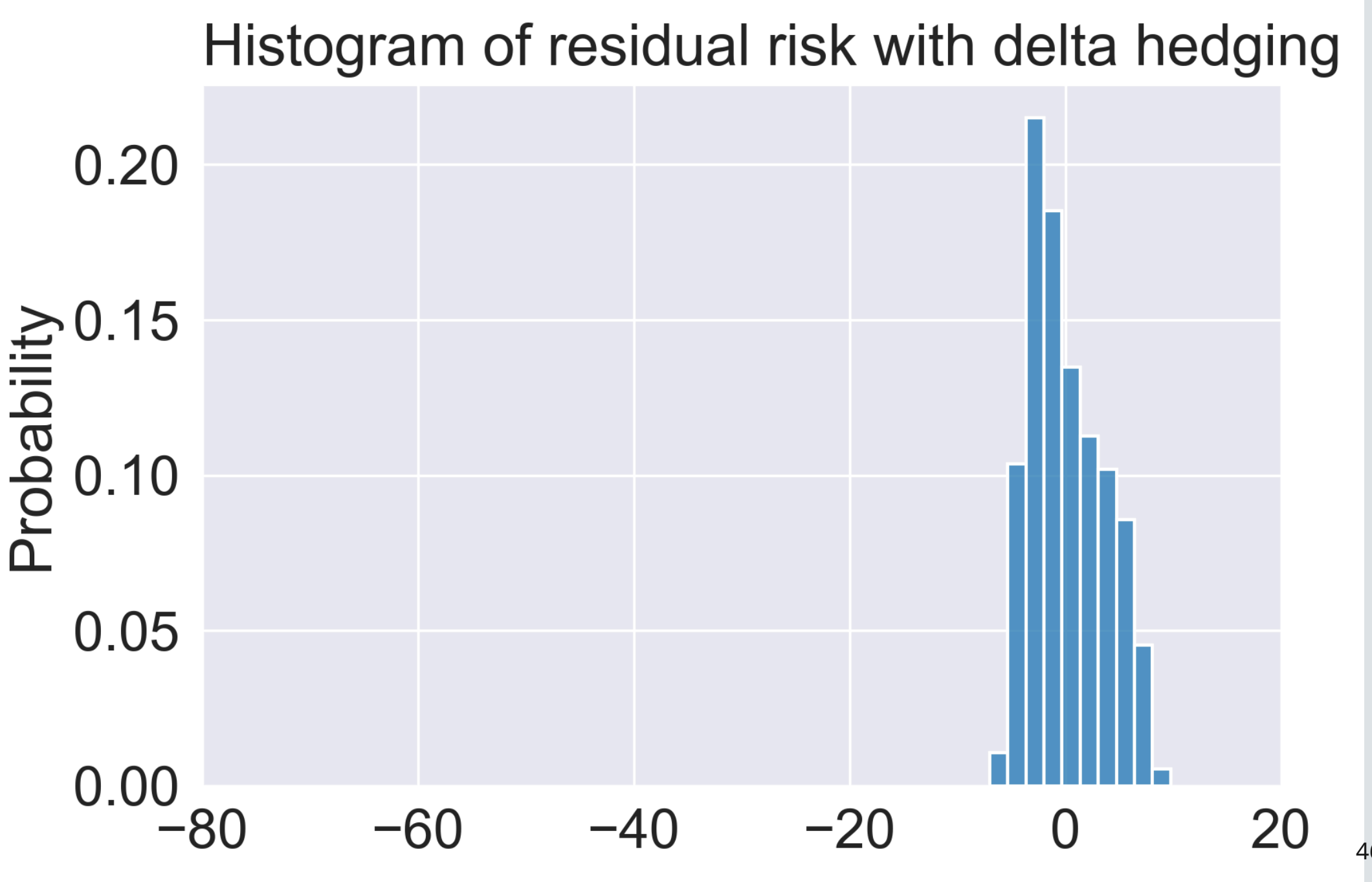
A Monte Carlo estimator of the option price $e^{-rT}\,\mathbb{E}_{\mathbb{Q}}[f(S_T)]$ is given by

$$\frac{1}{M}\sum_{m=1}^{M} e^{-rT} f(S_T^{\mathbb{Q},m})$$

where $S_T^{\mathbb{Q},m}$ are simulated under $\mathbb{Q}$ ($dS_t^{\mathbb{Q}} = rS_t^{\mathbb{Q}}dt + \sigma S_t^{\mathbb{Q}}dW_t$).
The idea of **Hedged Monte Carlo** is that this price should be equal to the one obtained under $\mathbb{P}$ ($dS_t^{\mathbb{P}} = \mu S_t^{\mathbb{P}}dt + \sigma S_t^{\mathbb{P}}dW_t$) **plus the simulation of the hedging strategy**, which gives an estimator with much lower variance

$$\frac{1}{M}\sum_{m=1}^{M}\left(e^{-rT} f(S_T^{\mathbb{P},m}) + \sum_{n=0}^{N}\Delta_{t_n}(S_{t_n}^{\mathbb{P},m})\times(S_{t_{n+1}}^{\mathbb{P},m} - S_{t_n}^{\mathbb{P},m})\right)$$


Histogram of residual risk without hedging
Probability
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20


Histogram of residual risk with delta hedging
Probability
0.20
0.15
0.10
0.05
0.00
−80
−60
−40
−20
0
20

Hedged Monte Carlo

One can push the idea further and directly optimize the option price P_0 and hedging strategy $\Delta_{t_n}(.)$ which minimize the average residual risk functional

$$\min_{P_0,\, \Delta_{t_n}(.), n=0,1,\ldots,N-1} \frac{1}{M} \sum_{m=1}^{M} \ell\left(P_0 - e^{-rT} f(S_T^{\mathbb{P},m}) - \sum_{n=0}^{N} \Delta_{t_n}(S_{t_n}^{\mathbb{P},m}) \times (S_{t_{n+1}}^{\mathbb{P},m} - S_{t_n}^{\mathbb{P},m}) \right)$$

where the transform ℓ is a loss function such as $\ell(x) = x^2$. This provides the **best empirical hedging strategy for any model** for the asset price. This optimal hedging is generally different from the Black-Scholes delta-hedging strategy.

Hedged Monte Carlo

More information about Hedged Monte Carlo is available in the following references:

[1] M. Potters, J. Bouchaud, and D. Sestovic. "Hedged Monte-Carlo: low variance derivative pricing with objective probabilities". In: *Physica A: Statistical Mechanics and its Applications* 289.3-4 (2001), pp. 517-525.

[2] J. Bouchaud and M. Potters. *Theory of financial risk and derivative pricing: from statistical physics to risk management*. 2nd. Cambridge University Press, 2003.